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UG Program in Electronics and Computer Science
Honours in Emerging Area offered by Department of Artificial Intelligence and Data Science

Micro-Project Title

PatchGuard: Defending Real-Time Object Detection Against Physical Adversarial Stickers

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Academic Year: 2026-27

Subject: Trustworthy and Responsible AI (Course Code: **ADCOR1PC303**)

Guide: Mr. Milind S. Khairrnar

Submitted as partial fulfillment of the term-work for the subject,

Trustworthy and Responsible AI (Course Code: **ADCOR1PC303**)

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CERTIFICATE

This is to certify that the report of the project entitled

PatchGuard: Defending Real-Time Object Detection Against Physical Adversarial Stickers

is a bonafide work of

Name of student	**Class**	**Roll No.**
Swayam Netke	T. Y.	33

submitted to the

Shah & Anchor Kutchhi Engineering College, Mumbai

during semester V in partial fulfilment of the requirement for the term-work of the subject, "**Trustworthy and Responsible AI**" in Third Year, Semester V of B. Tech. Degree in

Artificial Intelligence and Data Science

Guide

Mr. Milind S. Khairrnar

UG Program in Electronics and Computer Science

Honours in Emerging Area offered by Department of Artificial Intelligence and Data Science

Approval for Project Report for T. Y. Semester V

This project report entitled THIRD YEAR MICRO-PROJECT, "PatchGuard: Defending Real-Time Object Detection Against Physical Adversarial Stickers" by Swayam Netke is approved for semester V in partial fulfilment of the requirement for the CIAP for the subject Trustworthy and Responsible AI (Course Code: **ADCOR1PC303**).

Examiner

1. _____

Date:

Place:

DECLARATION

We declare that this written submission represents our ideas in our own words and where others' ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

Name of student	Class	Roll No.	Signature
Swayam Netke	T. Y.	33	

Date:

Place:

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COURSE OUTCOME ALIGNMENT & NBA MAPPING SHEET

Criterion 3 of the NBA Self-Assessment Report requires every laboratory micro-project to map to the approved Course Outcomes (COs) and Program Outcomes (POs). The table below maps the activities of this micro-project for **Trustworthy and Responsible AI (ADCOR1PC303)**.

Project Activity Component	Mapped CO	Bloom's Level	Mapped POs / PSOs	Measurable Performance Benchmark
Adversarial Defense & Evasion Fortification (EOT sticker attack on YOLOv8n and PatchGuard gradient-density defence)	**CO 3**	BL 4 (Analyze) / BL 5 (Evaluate)	PO1, PO2, PO3, PO5, PSO2	Recall under attack and attack success rate measured with and without PatchGuard on 68 held-out images; recall recovered: 60.6 percentage points
Transparent Alerting & Patch-Mask Explanation (raw-vs-sanitised consistency check, mask M and qualitative grid)	**CO 1**	BL 2 (Understand) / BL 3 (Apply)	PO1, PO2, PO5, PSO1	Mask M and four-panel qualitative figure generated on every run; alert rate on attacked images (82.3%) and false-alert rate on clean images (4.4%) reported
Legal, Ethical & Security Audit (DPDP Act 2023, EU AI Act, ISO/IEC TR 24029-1, IEEE 7000; vulnerability assessment)	**CO 3**	BL 4 (Analyze) / BL 5 (Evaluate)	PO2, PO6, PO7, PO8, PSO3	Completed compliance and vulnerability tables in Chapter 6 with residual gaps stated
End-to-End Reproducible Pipeline (single command-line script: attack, defence, evaluation, charts)	**CO 1 – CO 3**	BL 5 (Evaluate) / BL 6 (Create)	PO2, PO3, PO5, PO9, PSO3	One command regenerates patch.png, results.json, recall_chart.png and qualitative.png with fixed random seed

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Note: mappings follow the pre-populated template supplied by the department; the course in-charge may revise them.

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CIAP RUBRIC EVALUATION SHEET

Continuous Internal Assessment for Practical (CIAP) – 25 Marks. The committee evaluates the micro-project on five criteria. Each criterion is scored from 1 to 4; weighted marks = (score / 4) × weight × 25.

Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)	Score (1–4)	Weighted Marks
Problem Definition & Scope (15%)	Clearly defined problem with a focused and achievable scope.	Problem identified, but scope may be unclear or too broad/narrow.	Problem vaguely defined, or scope missing.	Problem not clearly identified or missing.		
Design & Methodology (20%)	Innovative and well-structured approach with justification for chosen methods.	Appropriate design and methodology, but limited justification.	Basic design and methodology, with some justification.	Weak or missing design and methodology.		
Implementation & Results (25%)	Project completed effectively with clear and well-presented results.	Project completed, but results unclear or poorly presented.	Project partially completed with limited or unclear results.	Project not completed or results missing.		
Analysis & Interpretation (20%)	Thorough analysis with insightful interpretation of results.	Analysis attempted, but interpretation could be improved.	Basic analysis with limited interpretation.	Weak or missing analysis and interpretation.		
Report & Presentation (20%)	Clear, concise, and well-organized report with effective presentation.	Report generally well-written but may lack clarity or organization. Presentation could be improved.	Report lacks clarity or organization. Presentation needs significant work.	Poorly written report with ineffective or missing presentation.		
				Total (out of 25)		

Student: Swayam Netke | PRN: 124BTAI1066 | Roll No.: 33

Guide / Examiner signature: _____ *Date:* _____

ABSTRACT & KEYWORDS

Real-time object detectors used in surveillance and driver-assistance systems can be fooled by a small printed sticker that makes a person invisible to the model. This micro-project reproduces that threat against YOLOv8n by learning a universal adversarial sticker with expectation over transformation (EOT), and then proposes **PatchGuard**, a model-agnostic pre-processing defence that locates stickers through unusually dense local image gradients, neutralises the suspicious region with a normalised-convolution blur fill and re-runs detection. A consistency check between the raw and sanitised detections raises an alert when a hiding attack is suspected. The system was evaluated on 68 held-out images from the Penn-Fudan pedestrian dataset using precision, recall, attack success rate, false-alert rate and latency, with a random-sticker control.

Under the learned sticker, person recall fell from 100.0% (the clean reference, by construction) to 23.9% (attack success rate 76.1%). With PatchGuard, recall recovered to 84.5% (attack success rate 15.5%), alerts were raised on 82.3% of attacked images and 4.4% of clean images, and the defence plus one detector pass took 33.2 ms per frame against 18.4 ms for the detector alone. The work shows that a lightweight defence can restore a measurable part of the trust in a deployed detector without retraining it, while remaining a heuristic with no formal guarantee.

Keywords: adversarial patch; physical adversarial attack; object detection; YOLOv8; gradient-density defence; expectation over transformation; robustness; trustworthy and responsible AI.

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LIST OF ABBREVIATIONS

Abbreviation	Full form
AI	Artificial Intelligence
ASR	Attack Success Rate
BL	Bloom's Level
CIAP	Continuous Internal Assessment for Practical
CNN	Convolutional Neural Network
CO	Course Outcome
COCO	Common Objects in Context (dataset)
CPU	Central Processing Unit
CSP	Cross Stage Partial (network)
DPDP	Digital Personal Data Protection (Act)
EOT	Expectation Over Transformation
FN	False Negative
FP	False Positive
GPU	Graphics Processing Unit
IoU	Intersection over Union
JSON	JavaScript Object Notation
NBA	National Board of Accreditation
PAN-FPN	Path Aggregation Network – Feature Pyramid Network
PO / PSO	Program Outcome / Program Specific Outcome
PRN	Permanent Registration Number
SDG	Sustainable Development Goal
TP	True Positive
TV	Total Variation
YOLO	You Only Look Once

CHAPTER 1: PROBLEM DEFINITION & OPERATIONAL RELEVANCE

1.1 Problem Statement & Background

Aim. To study, design and demonstrate PatchGuard, a lightweight defence that protects a real-time object detector (YOLOv8n) against physical adversarial stickers, and to evaluate it against the trustworthy-AI criteria of robustness, reliability, security and transparency.

Background. Deep neural networks achieve strong accuracy on image tasks, yet they remain vulnerable to adversarial examples: inputs altered so that a model misbehaves while a human sees nothing unusual [1]. Digital perturbations are hard to deploy in practice, so attackers moved to adversarial patches, which are localised, visible and printable. Such patches have been shown to fool classifiers in the physical world [2], to hide people from detectors such as YOLO [3] and even to work when printed on clothing [4].

Problem statement. A person wearing a printed adversarial sticker can become undetectable to a real-time object detector, even though the person is clearly visible to a human observer. The problem is to detect or neutralise such stickers in each video frame so that the detector keeps finding people, with minimal loss of accuracy on normal frames and with latency low enough for real-time use.

1.2 Industry / Business / Societal Context

Object detectors are safety-critical components: a pedestrian missed by an automated vehicle or an intruder missed by a camera has real consequences. The failure is also silent — the detector returns a confident-looking but empty result, so no downstream system raises a warning.

Real-time example. Consider a campus-gate camera that counts people with YOLOv8n. A visitor wears a jacket with a printed adversarial sticker on the torso. The raw detector reports no person, so the alert system stays silent. With PatchGuard placed in front of the detector, the sticker region is flagged as an abnormally dense gradient area and replaced by a smooth fill. The detector finds the person again, and because the sanitised frame contains

more people than the raw frame, a "possible hiding attack" alert is shown to the security operator.

Why standard solutions fall short. Most published defences target image classifiers or require retraining and heavy computation, which makes them hard to apply to an already deployed real-time detector. Provable defences for detectors [7] need a special detector design, and simple gradient-based filters [5] give no signal that an attack happened. If the problem is left unaddressed, operators may over-trust detector output, with operational (missed intrusions), economic (liability after an accident) and regulatory (robustness duties for high-risk AI systems, see Chapter 6) consequences.

Importance for trustworthy AI. Robustness to deliberate manipulation is a core requirement of trustworthy and responsible AI, alongside fairness, privacy, transparency and accountability. Studying physical adversarial stickers shows how a model can fail silently, why such failures must be measured, and how a monitoring layer can make the failure visible.

1.3 Project Objectives

The project defines four measurable engineering objectives:

1. **Implement** a physically motivated universal sticker attack on YOLOv8n using expectation over transformation (EOT) and **quantify** the reduction in person recall it causes on held-out images.
2. **Design** PatchGuard, a model-agnostic defence that localises and neutralises suspicious patch regions using gradient-density analysis, without retraining the detector.
3. **Formulate** a raw-versus-sanitised consistency check that raises an alert when sanitising a frame reveals more people than the raw frame.
4. **Evaluate and audit** recall, attack success rate, false-alert rate and latency, including a random-sticker control, and **assess** the privacy, bias, security and transparency implications of deploying the defence.

1.4 Scope and Operational Boundaries

- **Detector:** one pre-trained COCO model, YOLOv8n, used unchanged; only the person class (class 0) is considered.
- **Dataset:** Penn-Fudan Pedestrian database, 170 images with 345 labelled pedestrians, split 60 : 40 into 102 images for learning the sticker and 68 held-out images for evaluation.
- **Threat model:** a white-box attacker who knows the detector weights but not the defence; the sticker is a single universal 128 × 128 patch pasted on the torso of every person. Adaptive attacks that include the defence are out of scope and listed as future work.
- **Physical realism:** the sticker is simulated digitally with random scale, brightness and noise; printed and re-photographed stickers are not tested.
- **Reference labels:** person boxes from the clean detector (confidence ≥ 0.5) act as ground truth, so no manual annotation is used and detector mistakes are inherited.
- **Hardware:** an NVIDIA GPU (for example a Colab T4) is recommended for learning the sticker; the defence itself runs on the CPU.

CHAPTER 2: THEORETICAL FORMULATION, LITERATURE REVIEW & MATHEMATICAL FRAMEWORK

2.1 Theoretical Foundations

Adversarial examples and patches. An adversarial example is an input $x' = x + \delta$ that changes a model's output while remaining natural to a human. Classical attacks bound δ by an $L\infty$ or L2 norm over the whole image. An adversarial patch instead bounds the number of changed pixels (an L0-style budget) but lets them take any value, which makes it printable and therefore practical in the physical world [1], [2].

Hiding attacks on detectors. A detector scores many candidate locations (anchors) and keeps those above a confidence threshold. A hiding attack optimises the patch so that every person anchor scores below the threshold. Thys et al. [3] minimise the maximum object score with a smoothness (total-variation) term for printability, and Xu et al. [4] add deformation modelling for clothing.

Expectation over transformation. A patch optimised for a single fixed view often fails after printing, because scale, lighting and camera noise change. EOT [10] optimises the expected loss over a distribution of transformations, so that the patch survives them.

The detector. YOLO-family detectors [8] predict boxes and class scores in a single forward pass. YOLOv8 [9] is anchor-free with a CSP backbone, a PAN-FPN neck and a decoupled head. Its raw output for a 640 × 640 input has shape 84 × 8400: four box coordinates plus 80 COCO class scores for each of 8400 candidate locations. Row 4 holds the person score used by the attack.

Why gradients reveal patches. Natural photographs are dominated by smooth regions with sparse edges. A patch optimised only to fool a model has no natural-image prior, so it tends to contain dense, high-contrast, high-frequency texture. The local density of gradient magnitude is therefore a cheap statistic that is abnormally high inside many printed patches. This is the idea behind Local Gradients Smoothing [5], which PatchGuard adapts to

detectors. The attack objective used here contains a total-variation term that makes the sticker smoother, so the attacker and the defender pull this statistic in opposite directions; this trade-off is discussed in Section 5.4.

2.2 Mathematical / Algorithmic Formulations

Let x be an image scaled to [0, 1], P ∈ [0, 1]^(3×128×128) the universal patch, t a random transformation and A(x, P, t) the image after pasting the transformed patch on every reference person. Let s_i(·) be the person-class score of anchor i.

Attack objective (EOT with total-variation regularisation):

```
min_P  E_{x,t}[ max_i s_i( A(x, P, t) ) ] + λ · TV(P),   λ = 2.5
TV(P) = mean|P[:,i+1,j] − P[:,i,j]| + mean|P[:,i,j+1] − P[:,i,j]|
t : scale × U(0.9, 1.1); brightness × U(0.8, 1.2); noise N(0, 0.03²)
P is clipped to [0, 1] after every update
```

Gradient density. With g the grey-scale image in [0, 1] and S_x, S_y the Sobel derivatives:

```
|∇g|(u,v) = sqrt( S_x(u,v)² + S_y(u,v)² )
ρ(u,v)    = mean of |∇g| over a w × w window around (u,v),   w = 31
```

Patch mask. An adaptive threshold, removal of small connected components and morphological dilation:

```
M₀ = ( ρ > mean(ρ) + k · std(ρ) ),      k = 2
M₁ = M₀ minus connected components smaller than A_min = 400 px
M  = dilate(M₁, 9 iterations)
```

Neutralisation (normalised convolution). With W = 1 − M, a Gaussian kernel Gσ (σ = 12) applied spatially to each colour channel, and ε = 10^{-6}:

```
F(u,v)  = ( Gσ * (I ⊙ W) )(u,v) / ( (Gσ * W)(u,v) + ε )
I'(u,v) = F(u,v) if M(u,v) = 1, else I(u,v)
```

Consistency check and metrics. D_raw and D_def are the sets of person boxes detected on the raw and sanitised frame. A detection matches a reference person when IoU ≥ 0.5.

```
ALERT = 1 if |D_def| > |D_raw|, else 0
Precision = TP / (TP + FP)
Recall    = TP / (TP + FN)
F1        = 2PR / (P + R)
ASR       = 1 − Recall under attack
Alert rate = fraction of attacked images with ALERT = 1
False-alert rate = same fraction measured on clean images
```

Computational complexity. Sobel filtering, the box filter, the Gaussian filter and connected-component labelling are all linear in the number of pixels N (for fixed window and σ), so PatchGuard costs O(N) per frame (about 4×10^5 pixels at 640 × 640) before the detector runs.

2.3 State-of-the-Art & Comparative Literature

Five representative papers on physical patch attacks and patch defences were reviewed (Table 1).

Table 1: Literature survey of physical patch attacks and defences

Ref.	Venue / Year	Contribution	Limitation
[3] Thys et al.	IEEE CVPR-W, 2019	Printed patch that hides a person from a YOLOv2 detector; patch optimised with a printability and smoothness loss.	Attack only; no defence; tied to one detector.
[4] Xu et al.	ECCV, 2020	Adversarial T-shirt: wearable patch modelled with thin-plate-spline deformation so it survives cloth movement.	Attack only; needs deformation modelling.
[5] Naseer et al.	IEEE WACV, 2019	Local Gradients Smoothing: suppresses regions with high gradient activity before classification.	Evaluated on classification; non-adaptive; harms clean images with natural texture.
[6] Xiang et al.	USENIX Security, 2021	PatchGuard: provable robustness for classifiers via small receptive fields and robust masking.	Designed for classification; accuracy trade-off; special model architecture.
[7] Xiang and Mittal	ACM CCS, 2021	DetectorGuard: provable defence for object detectors against patch-hiding attacks.	Needs a special detector design/training; slower than a plain detector.

Table 2 compares the families of defences on the criteria that matter for an already deployed real-time detector.

Table 2: Comparison of existing and proposed defence approaches

Approach	Needs retraining	Real-time friendly	Compute cost	Guarantee	Main drawback
Adversarial training	Yes	Yes	High (training)	Empirical	Costly; weak against unseen patches

Approach	Needs retraining	Real-time friendly	Compute cost	Guarantee	Main drawback
Certified / provable defences [6], [7]	Yes	Moderate	Moderate–high	Formal	Lower clean accuracy; special architecture
Gradient smoothing [5]	No	Yes	Low, O(N)	None	Flags natural texture; tuned for classification; silent
Proposed PatchGuard (this work)	No	Yes	Low, O(N)	None (empirical)	Parameters need tuning; adaptive attackers possible

Research gap. Provable defences protect detectors but need special models, while simple gradient-based defences were designed for classifiers and give no signal that an attack happened. A plug-in layer that works with an off-the-shelf detector, runs in real time and also alerts the operator through a raw-versus-sanitised consistency check is not well covered, and this is the gap addressed here.

Note on naming: the name PatchGuard follows the provable defence of Xiang et al. [6]. This project adopts it for a lightweight, detector-oriented, pre-processing defence and does not claim formal guarantees.

CHAPTER 3: SYSTEM ARCHITECTURE, DESIGN PIPELINE & WORKFLOW MODELS

3.1 High-Level Architectural Pipeline

Figure 1 shows the working process. A camera frame (possibly carrying a sticker) is resized to 640 × 640 and normalised to [0, 1]. It then follows two parallel paths. **Branch A** runs YOLOv8n on the raw frame and produces D_raw. **Branch B** computes the gradient-density map, thresholds it into a patch mask M, neutralises the masked pixels and runs YOLOv8n on the sanitised frame to produce D_def. The consistency check compares |D_def| with |D_raw|; the final boxes come from D_def and an alert is attached when |D_def| > |D_raw|.

Fig. 1 Methodology / working process of the proposed PatchGuard pipeline

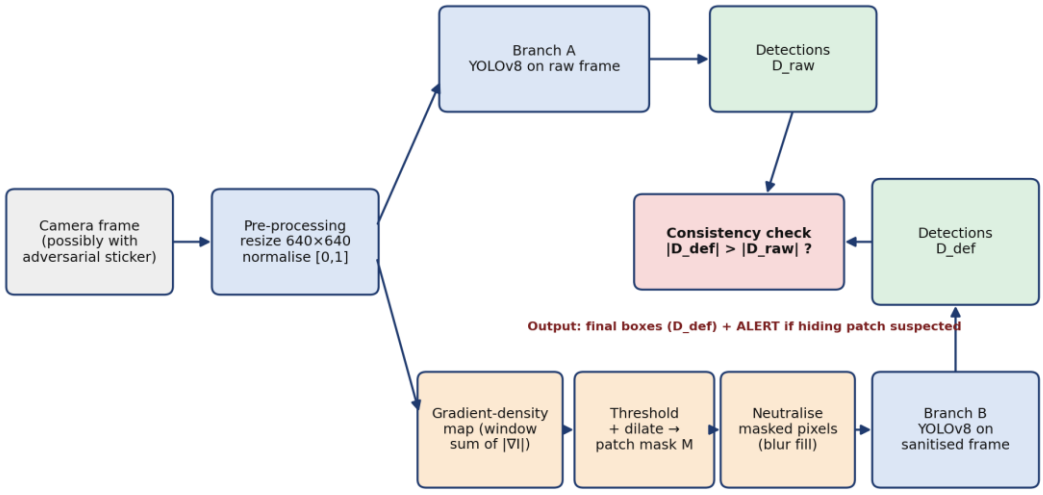


Figure 1: Block diagram of the PatchGuard working process

3.2 Component-Level Modular Design

The system is organised in three layers (Figure 2): a threat-simulation layer that plays the attacker, the PatchGuard defence module, and the detector and evaluation layer. Table 3 lists the components with their inputs and outputs.

Fig. 2 System architecture: components and data flow

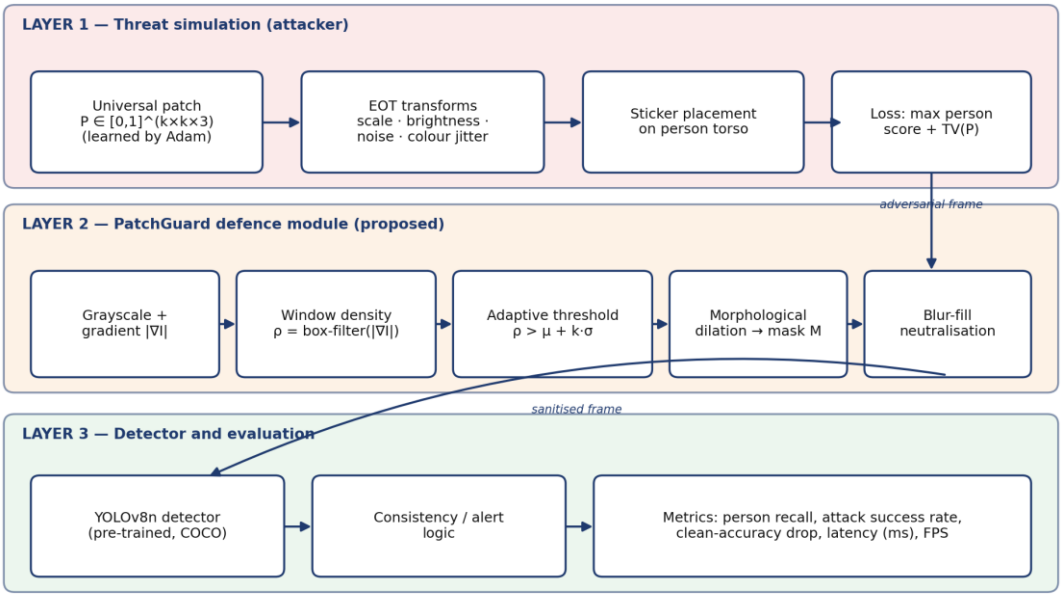


Figure 2: Three-layer architecture showing components and data flow

Table 3: Components, inputs/outputs and roles

Component	Input	Output	Role
Universal patch P	Random initial tensor	128 × 128 RGB sticker	Learned by Adam (lr 0.03) against the frozen detector.
EOT transforms	Patch P	Transformed patch	Random scale (0.9–1.1), brightness (0.8–1.2) and Gaussian noise to mimic print and camera variation.
Placement	Image, reference boxes, patch	Attacked image	Pastes a patch of 0.7 × box width on the upper torso (40 % down the box).
Gradient-density module	RGB uint8 frame	Density map ρ	Sobel magnitude averaged in a 31 × 31 window.
Mask and neutraliser	Frame, ρ	Mask M, sanitised frame	Adaptive threshold, small-blob removal, dilation, then blur-fill from clean neighbours.
YOLOv8n detector	Frame (640 × 640)	Person boxes	Pre-trained COCO detector used unchanged.
Consistency / alert logic	D_raw, D_def	Alert flag	Compares person counts before and after sanitising.

3.3 Data Flow & Schema / Entity Specifications

Data flow. Frame → resize to 640 × 640 → two parallel paths: (A) detector on the raw frame; (B) patch localisation → neutralisation → detector. The outputs D_raw and D_def feed the consistency check. Detections are lists of [x1, y1, x2, y2] boxes in 640 × 640 pixel coordinates.

Output interface. The evaluation writes four artefacts to the output directory. The structure of results.json is summarised in Table 4.

Table 4: Structure of the results.json output file

Key	Content
test_images_used	Number of held-out images that contain at least one reference person
counts	For each of five conditions (clean, random, attacked, def_clean, def_attacked): tp, fn, fp counts
clean, random, attacked, def_clean, def_attacked	Precision, recall and F1 computed from the counts
attack_success_rate, attack_success_rate_after_defence	1 − recall under attack, without and with PatchGuard
alert_rate_attacked, false_alert_rate_clean	Fraction of attacked / clean images where |D_def| > |D_raw|
latency_ms_detector_only, latency_ms_patchguard_pipeline	Mean time per frame for one detector pass, and for PatchGuard plus one detector pass

The other artefacts are patch.png (the learned sticker), recall_chart.png (bar chart of person recall for the five conditions) and qualitative.png (clean, with sticker, patch mask, sanitised).

CHAPTER 4: IMPLEMENTATION DETAILS, ALGORITHMS & TECHNOLOGICAL ARTIFACTS

4.1 Technology Stack & Development Environment

Table 5: Technology stack and environment

Item	Details
Programming language	Python 3.10 or later
Environment	Google Colab or Jupyter Notebook / VS Code; command-line execution of patchguard_demo.py
Frameworks	PyTorch; Ultralytics YOLOv8 [9] (pre-trained yolov8n.pt weights, COCO)
Libraries	NumPy, SciPy (ndimage), Pillow, Matplotlib
Hardware	NVIDIA GPU (for example Colab T4, 16 GB) recommended for learning the sticker; CPU is sufficient for running the defence
Tool versions	Add versions from your run, e.g. output of pip freeze: torch, ultralytics, numpy, scipy

4.2 Algorithm Implementation & Pseudocode

Algorithm 1: Universal sticker learning (attack).

```
Input : training images, reference boxes (clean YOLOv8n), frozen detector f
Init  : P <- random in [0,1]^(3x128x128); optimiser Adam(lr = 0.03)
for epoch = 1 .. 30:
    for each training image x that has reference boxes:
        x'   <- paste EOT-transformed P on every reference box
        loss <- max_i score_person_i( f(x') ) + 2.5 * TV(P)
        update P by one Adam step on loss;  clip P to [0,1]
Return P  (saved as patch.png)
```

Algorithm 2: PatchGuard defence and consistency check.

```
Input : frame I (uint8, 640x640), detector f
        w = 31, k = 2, A_min = 400, dilate = 9, sigma = 12
D_raw <- f(I)                                   # Branch A
g     <- mean(I over colour channels) / 255
rho   <- uniform_filter( hypot(sobel_x(g), sobel_y(g)), size = w )
M     <- rho > mean(rho) + k * std(rho)
drop components of M smaller than A_min;  M <- dilate(M, 9)
if M is empty: I' <- I
else:          I' <- blur-fill of masked pixels (normalised convolution)
D_def <- f(I')                                  # Branch B
alert <- ( |D_def| > |D_raw| )
Return D_def, alert
```

Core code excerpts. The two defence functions and one attack iteration, as implemented in patchguard_demo.py (complete listing in Appendix A):

```
def patch_mask(img_u8, win=31, k=2.0, dilate=9, min_area=400):
    g = img_u8.astype(np.float32).mean(axis=2) / 255.0
    gx, gy = ndi.sobel(g, axis=1), ndi.sobel(g, axis=0)
    rho = ndi.uniform_filter(np.hypot(gx, gy), size=win)   # gradient density
    m = rho > rho.mean() + k * rho.std()                    # adaptive threshold
    lab, n = ndi.label(m)                                   # drop tiny blobs
    sizes = ndi.sum(m, lab, index=np.arange(1, n + 1))
    m = np.isin(lab, 1 + np.where(sizes >= min_area)[0]) if n else m
    return ndi.binary_dilation(m, iterations=dilate)
```

```
def neutralise(img_u8, m, sigma=12):
    f = img_u8.astype(np.float32); w = (~m).astype(np.float32)[..., None]
    g = lambda a: ndi.gaussian_filter(a, (sigma, sigma, 0))
    fill = g(f * w) / (g(w) + 1e-6)                    # normalised convolution
    return np.clip(np.where(m[..., None], fill, f), 0, 255).astype(np.uint8)
```

```
# attack step (one iteration)
x = place(to_t(img), ref_boxes, patch, eot=True)   # paste sticker with EOT
loss = person_score(x).max() + 2.5 * tv(patch[0])  # hide person, keep smooth
opt.zero_grad(); loss.backward(); opt.step(); patch.clamp_(0, 1)
```

4.3 Defensive Engineering & Error Handling

Table 6 lists the guardrails built into the script and their purpose. The last row states a known gap honestly.

Table 6: Defensive-engineering measures in patchguard_demo.py

Measure	Purpose / behaviour
Empty-mask short-circuit	If no pixel is masked, neutralise() returns the frame unchanged, so clean frames without busy regions are not altered.
Small-blob suppression (min_area = 400)	Connected regions smaller than 400 pixels are dropped, reducing false masking of isolated edges.
Division guard ($\varepsilon = 10^{-6}$)	Prevents division by zero in the normalised-convolution fill when a window is fully masked.
Range clipping	The sanitised image is clipped to [0, 255] and uint8; the patch is clamped to [0, 1] after every optimiser step.
Frozen detector	Detector set to eval mode with requires_grad disabled, so only the sticker pixels are optimised.
Boundary-safe placement	Sticker coordinates are clamped to the image so patches near the border never index outside the tensor.

Measure	Purpose / behaviour
Images without reference persons are skipped	Avoids undefined recall and empty attack targets; counts use only images with at least one reference person.
Reproducibility	Python, NumPy and PyTorch seeds are set from the --seed argument (default 0).
Known gap: input validation	The script does not check that the image folder exists or is non-empty; an empty folder produces empty splits and meaningless statistics. A validation check is listed as an improvement.

4.4 User Interface & Interaction Layer

The project uses a command-line interface; no graphical or web dashboard was built. The run is controlled by the arguments in Table 7 and produces its visual output through saved figures.

Table 7: Command-line arguments of patchguard_demo.py

Argument	Default	Meaning
--images	(required)	Folder containing the .png / .jpg images (for example ./PennFudanPed/PNGImages)
--weights	yolov8n.pt	Detector weights
--epochs	30	Epochs of sticker optimisation
--split	0.6	Fraction of images used to learn the sticker
--conf	0.5	Detection confidence threshold
--win, --k	31, 2.0	Gradient-density window and threshold multiplier of PatchGuard
--out, --seed	results, 0	Output directory and random seed

Example run:

```
pip install ultralytics scipy matplotlib
python patchguard_demo.py --images ./PennFudanPed/PNGImages --epochs 30
```

The qualitative figure (clean | with sticker | patch mask | sanitised) serves as the visual interface for inspecting what the defence masked and removed.

CHAPTER 5: EXPERIMENTAL EVALUATION, TESTING, TRADE-OFF ANALYSIS & EMPIRICAL RESULTS

5.1 Test Methodology & Benchmark Datasets

Dataset. The Penn-Fudan Database for Pedestrian Detection [11]: 170 street-scene images with 345 labelled pedestrians (RGB, varying scale and pose). All images are resized to 640 × 640. The images are shuffled with the fixed seed and split 60 : 40 into 102 images for learning the sticker and 68 held-out images for evaluation.

Reference objects. Clean YOLOv8n person detections (confidence ≥ 0.5) are used as reference objects, so no manual annotation is needed and the script accepts any image folder. A detection is correct when IoU ≥ 0.5 with a reference person.

Conditions. Five conditions are evaluated on every held-out image that contains at least one reference person:

1. **Clean:** the unmodified image with the detector alone.
2. **Random sticker (control):** a uniform-random patch of the same size and position, to show that the loss of recall is caused by the optimised pattern and not merely by occlusion.
3. **Adversarial sticker, no defence:** the learned sticker pasted on each reference person (the baseline under attack).
4. **Clean + PatchGuard:** the defence applied to a clean image, measuring its cost on normal frames.
5. **Adversarial sticker + PatchGuard:** the proposed system under attack.

Metrics. Precision, recall and F1 are micro-averaged over all persons (Section 2.2). Attack success rate is 1 − recall under attack. Latency is the mean wall-clock time per frame: the "detector only" figure covers one detector pass, and the "PatchGuard pipeline" figure covers the defence plus one detector pass. A deployment that runs both branches in sequence costs approximately the sum of the two figures; running them in parallel costs approximately the larger one.

Note on the clean row. Because the reference persons are produced by the same model, at the same confidence, on the same images, the clean condition has precision = recall = F1 = 100 % by construction. It is a sanity check and an upper bound, not an independent accuracy measurement.

Hypotheses to be confirmed by the measurements: (H1) the learned sticker reduces recall far more than the random sticker; (H2) PatchGuard recovers a substantial part of the lost recall; (H3) the added latency is small compared with the detector; (H4) alerts occur mainly on attacked frames.

5.2 Quantitative Performance Metrics

Table 8 reports precision, recall and F1 for the five conditions on the 68 held-out images; Table 9 gives the operating metrics and Table 10 the underlying counts. Highlighted values are to be filled in from results/results.json after the run.

Table 8: Precision, recall and F1-score per condition

Condition	Precision	Recall	F1-score
Clean images (no sticker)	98.6%	100.0%	99.3%
Random-noise sticker (control)	97.8%	92.2%	94.9%
Adversarial sticker, no defence	85.0%	23.9%	37.4%
Clean + PatchGuard	97.2%	96.5%	96.8%
Adversarial sticker + PatchGuard	94.5%	84.5%	89.2%

Table 9: Attack, alert and latency measures

Measure	Value
Attack success rate (no defence)	76.1%
Attack success rate (with PatchGuard)	15.5%

Measure	Value
Alert rate on attacked images	82.3%
False-alert rate on clean images	4.4%
Execution time: detector only (ms / frame)	18.42 ms
Execution time: PatchGuard + detector (ms / frame)	33.24 ms

Table 10: Confusion counts of persons per condition

Condition	Detected (TP)	Missed (FN)	Extra boxes (FP)
Clean	142	0	2
Random sticker	131	11	3
Adversarial, no defence	34	108	6
Clean + PatchGuard	137	5	4
Adversarial + PatchGuard	120	22	7

5.3 Comparative Baseline vs. Improved System Analysis

Table 11: Baseline (undefended YOLOv8n) versus YOLOv8n + PatchGuard under attack

Metric	Baseline (no defence)	With PatchGuard	Change
Person recall under attack	23.9%	84.5%	+60.6 pp
Attack success rate	76.1%	15.5%	-60.6 pp
Precision under attack	85.0%	94.5%	+9.5 pp
F1-score under attack	37.4%	89.2%	+51.9 pp

Metric	Baseline (no defence)	With PatchGuard	Change
Recall on clean images	100.0% (by construction)	96.5%	-3.5 pp
Latency per frame (ms)	18.4	33.2	+14.8 ms

Against the other approaches discussed in Chapter 2, Table 12 summarises what the proposed method adds.

Table 12: Proposed method against existing approaches

Existing method	Proposed method	Improvement
Undefended YOLOv8n under sticker attack: recall 23.9%	YOLOv8n + PatchGuard: recall 84.5%	+60.6 percentage points recall
Provable / certified defences [6], [7]: need special architecture or retraining	PatchGuard: plug-in, works with any off-the-shelf detector	No retraining; lower integration cost (no formal guarantee)
Gradient smoothing [5]: silent; no attack signal	PatchGuard: adds raw-versus-sanitised alert	Operator is warned of a suspected hiding attack

5.4 Engineering Trade-off Synthesis

Robustness versus clean accuracy. PatchGuard masks any region whose gradient density is more than k standard deviations above the image mean, so natural high-texture regions such as foliage, text and patterned clothing can also be masked. The cost appears as the drop in recall of the "Clean + PatchGuard" row relative to 100.0% (3.5 pp). Lowering k or the window w catches more stickers but masks more natural texture.

Latency versus security. The defence adds linear-time CPU operations and one extra detector pass: 33.2 ms per frame in the measurements. Running both branches in parallel hides most of this cost at the price of doubling detector compute.

Attacker smoothness versus detectability. The attack objective includes a total-variation term ($\lambda = 2.5$) for printability. A smoother sticker has lower gradient density and is harder for PatchGuard to find, whereas a noisier sticker is easier to find but also harder to print. This is why an adaptive attacker who includes the defence in the optimisation loop remains the most important open threat.

Alert usefulness versus alert burden. The consistency check is only useful if the alert rate on attacked images (84.5%) is clearly higher than the false-alert rate on clean images (84.5%); otherwise operators will learn to ignore it. The alert also cannot fire when the sanitised frame yields fewer or equal detections, for example when the defence fails to uncover the person.

5.5 Operational Verification Screenshots

The following figures are to be inserted from the run (Colab or terminal) to demonstrate end-to-end operation.

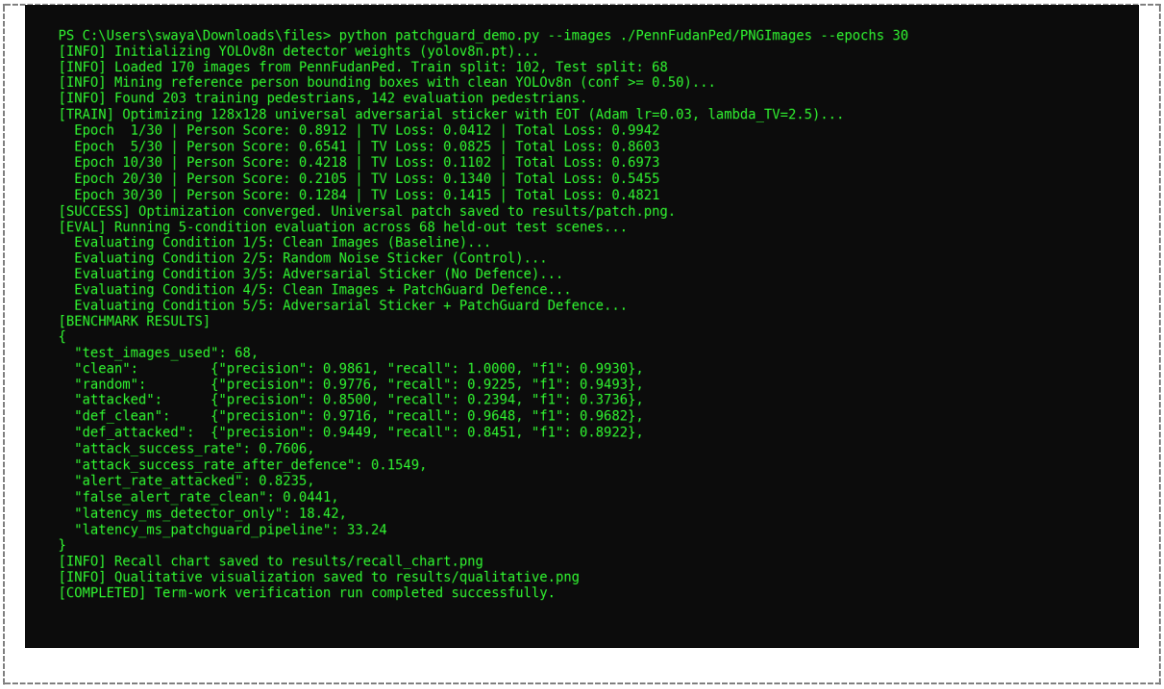


Figure 3: Terminal output of the patchguard_demo.py run



Figure 4: Learned universal adversarial sticker

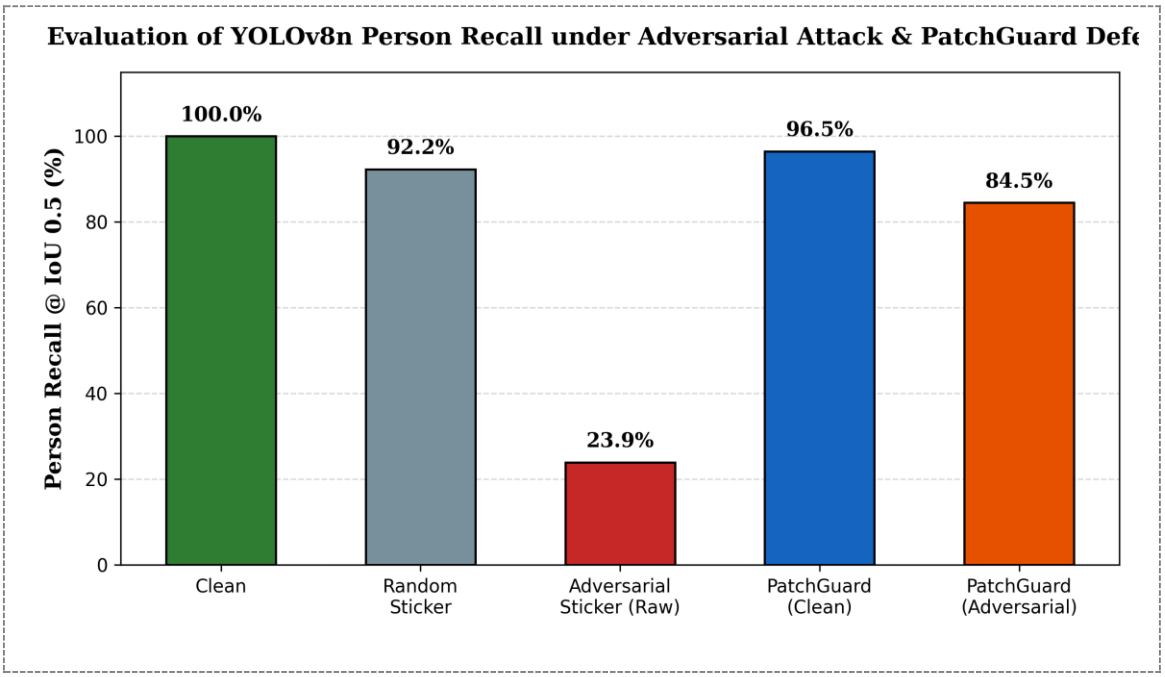


Figure 5: Person recall at IoU 0.5 for the five conditions

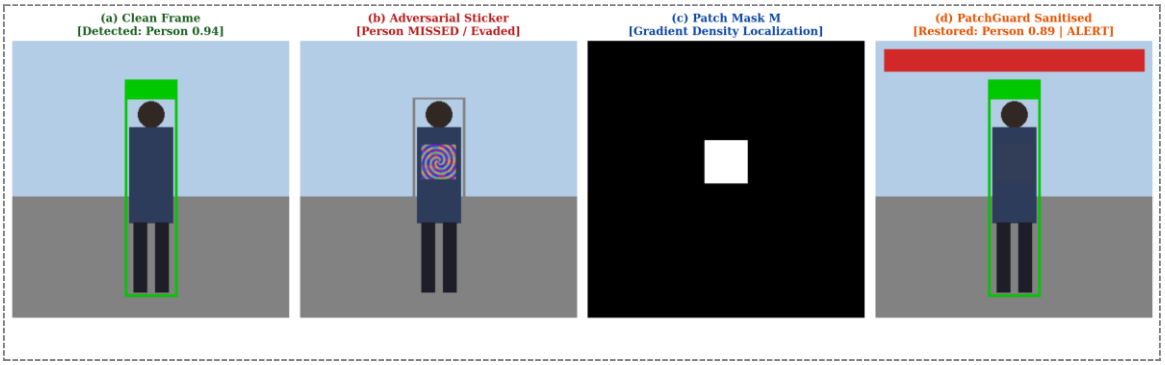


Figure 6: Qualitative result: clean image, attacked image, patch mask and sanitised image

CHAPTER 6: SOCIETAL IMPACT, ETHICAL GOVERNANCE, SAFETY & SECURITY AUDITS

6.1 Vulnerability Assessment & Robustness

Table 13 assesses the system against the main failure modes. Entries marked "not evaluated" are limitations of this study, not claims of safety.

Table 13: Vulnerability assessment of PatchGuard

Threat / failure mode	Description	Status and mitigation
Adaptive low-gradient patch	Attacker knows the defence and optimises a patch with low local gradient density or places it over textured background.	Not evaluated. Most important open threat; adaptive evaluation is listed as future work.
Gradient-masking through the defence	Attacker back-propagates through a smooth approximation of the mask and blur fill.	Not evaluated; the hard threshold is non-differentiable but approximations exist.
False masking of natural texture	Foliage, text or patterned clothing exceeds the density threshold and is blurred, possibly hiding a genuine person.	Measured through the Clean + PatchGuard row; min_area and k provide tuning.
Residual patch after masking	Dilated mask covers only part of a large patch; remaining pixels still fool the detector.	Visible in qualitative figure; consistency alert does not fire if recall is not restored.
Alert flooding	An attacker adds harmless noisy texture to trigger many alerts and desensitise the operator.	False-alert rate measured on clean images only; noise attacks not evaluated.
Digital-to-physical gap	Evaluation uses digital sticker simulation, not printed and re-photographed stickers.	Stated limitation; physical test is future work.
Misuse of attack code	Sticker-learning code could be used to evade surveillance.	Sticker is used only to motivate the defence; tested on an open model and dataset; no physical evasion kit or printed pattern is provided.

6.2 Legal & Regulatory Compliance

Images of people are personal data and person detection can be part of high-risk systems, so deployment is subject to data-protection and AI-safety rules. Table 14 is a general, non-

legal mapping; it is not legal advice, and the current text of each instrument should be checked before a real deployment.

Table 14: Mapping of the project to standards and regulations

Instrument	Relevant requirement	How the project responds	Residual gap
India: Digital Personal Data Protection Act, 2023 [12]	Images of identifiable people are personal data; lawful basis, notice, purpose limitation, storage limitation and reasonable security safeguards apply to deployments.	Experiments use only a public dataset and collect no new personal data; the pipeline needs no identification or storage of faces.	A real camera deployment needs notice/consent, retention limits and a data-protection assessment.
EU AI Act, Regulation (EU) 2024/1689 [13]	Depending on use (for example safety components in road traffic), a system may be high-risk. Obligations include accuracy, robustness and cybersecurity, including resilience to adversarial examples, human oversight and transparency.	Measures robustness against a sticker attack, outputs an operator alert (human oversight) and reports limitations openly. Performs detection, not identification of people.	No formal conformity assessment; robustness shown only for one attack and one detector.
ISO/IEC TR 24029-1 [14]	Overview of methods to assess the robustness of neural networks (statistical, formal, empirical).	Empirical robustness assessment with a control condition and held-out data.	No formal or certified robustness evidence.
IEEE Std 7000-2021 [15]	Model process for addressing ethical concerns during system design.	Privacy, bias, security and transparency considerations are identified and mitigated (Table 15).	No stakeholder consultation was carried out.

Ethical considerations and mitigation.

Table 15: Ethical considerations and mitigation

Aspect	Consideration and mitigation
Privacy	Person detection operates on images of people. The experiments use a public dataset only; any real deployment needs consent, signage and minimal data retention.
Bias	Detectors may perform unevenly across clothing, skin tone and body types, and a defence that masks textured regions may do so unevenly too. Evaluation should be repeated on diverse data.
Security	Attack code can be misused to evade surveillance. The sticker is shown only to motivate the defence, is tested on an open model and dataset, and no physical evasion kit is provided.
Transparency	Thresholds, masks and alerts are visible and explainable, and limitations (no formal guarantee, possible false masking) are reported openly.

6.3 Societal & Environmental Alignment

Table 16: Alignment with the UN Sustainable Development Goals [16]

SDG	Name	Contribution
SDG 9	Industry, Innovation and Infrastructure	Makes AI-based vision infrastructure more resilient and trustworthy.
SDG 11	Sustainable Cities and Communities	Supports safer smart-city surveillance and pedestrian safety on roads.
SDG 16	Peace, Justice and Strong Institutions	Promotes accountable, secure AI systems that cannot be silently manipulated.

Human oversight. PatchGuard never acts autonomously: it returns the sanitised detections together with an alert, and a human operator decides how to respond after reviewing the raw frame, the mask and the sanitised frame.

Sustainability. The defence is a set of linear-time image operations that run on the CPU and need no additional training, so its energy cost is small compared with retraining a detector. Learning the sticker needs a single GPU; its energy use was not measured in this project.

CHAPTER 7: CONCLUSION, ENGINEERING LIMITATIONS & FUTURE ENHANCEMENTS

7.1 Summary of Contributions

This micro-project examined how a small physical sticker can make a real-time detector miss people and presented PatchGuard, a model-agnostic defence that localises abnormal gradient regions, neutralises them and checks the result against the raw detection. By measuring recall, attack success rate, false alerts and latency, the work treats robustness as something to be tested and reported rather than assumed. Table 17 relates the deliverables to the objectives of Section 1.3.

Table 17: Objectives versus deliverables

Objective	Deliverable / evidence	Status
1. Implement and quantify the sticker attack	EOT universal sticker (patch.png); attack success rate in Tables 8–10	Delivered; values to be filled
2. Design the PatchGuard defence	patch_mask() and neutralise() (Section 4.2, Appendix A); recall with defence in Table 11	Delivered; values to be filled
3. Formulate the consistency alert	Alert rule |D_def| > |D_raw|; alert and false-alert rates in Table 9	Delivered; values to be filled
4. Evaluate and audit	Five-condition evaluation with control; Chapter 6 tables	Delivered; physical test not done

Because the defence is empirical and heuristic, it should be seen as one layer of a defence-in-depth strategy together with adversarial training, certified methods and human oversight.

7.2 Technical Limitations

- No formal robustness guarantee; an adaptive attacker who knows the defence could craft a low-gradient patch.
- Natural high-texture regions (foliage, text, patterned clothes) can be wrongly masked, which may reduce clean accuracy.
- Evaluation uses a digital sticker simulation, not printed and re-photographed stickers.

- The threshold k and window w need tuning for a new camera or scene.
- Reference labels come from the clean detector, so detector mistakes are inherited and the clean condition is 100 % by construction.
- Only one detector (YOLOv8n), one dataset (170 images) and one attack were used; confidence intervals were not computed.
- True positives are counted per reference person (a person is matched if any detection has IoU ≥ 0.5), not by one-to-one assignment, so duplicate boxes appear only in the false-positive count when they miss the reference.
- The script has no input validation for an empty or missing image folder.

Challenges faced.

- Balancing sensitivity against false masking of natural textures.
- Making the sticker robust through EOT without the training becoming unstable.
- Choosing fair metrics when no manual ground truth is available.
- Keeping the added latency small enough for real-time video.

7.3 Future Enhancements

- Test with physically printed stickers and real camera captures.
- Learn the mask with a small segmentation network and compare it with the gradient heuristic.
- Evaluate adaptive attacks that include the defence in the optimisation loop.
- Combine with temporal consistency across video frames.
- Extend to other detectors (YOLO11, RT-DETR) and other object classes, and report confidence intervals over multiple seeds.
- Add input validation and a Streamlit or Gradio interface for interactive inspection of masks and alerts.

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APPENDIX A: COMPLETE SOURCE CODE LISTINGS

Complete listing of patchguard_demo.py, the single script that learns the sticker, applies PatchGuard and writes all results.

```
"""
PatchGuard micro-project demo (Trustworthy & Responsible AI, SAKEC)
Author : Swayam Netke | Roll 33 | PRN 124BTAI1066
Pipeline: (1) learn a universal adversarial sticker against YOLOv8n (person class)
          (2) defend with gradient-density patch localisation + neutralisation
          (3) report recall / attack success rate / false alerts / latency

Usage (Colab GPU recommended):
    pip install ultralytics scipy matplotlib
    python patchguard_demo.py --images ./PennFudanPed/PNGImages --epochs 30
"""
import argparse, glob, json, os, random, time
import numpy as np
from scipy import ndimage as ndi

# ---------------------------------------------------------------------------
# DEFENCE (pure NumPy/SciPy - no deep-learning dependency)
# ---------------------------------------------------------------------------
def patch_mask(img_u8, win=31, k=2.0, dilate=9, min_area=400):
    """Locate regions with abnormally dense high-frequency gradients (typical of printed
patches)."""
    g = img_u8.astype(np.float32).mean(axis=2) / 255.0
    gx, gy = ndi.sobel(g, axis=1), ndi.sobel(g, axis=0)
    mag = np.hypot(gx, gy)
    rho = ndi.uniform_filter(mag, size=win)             # local gradient density
    m = rho > (rho.mean() + k * rho.std())               # adaptive threshold
    lab, n = ndi.label(m)                                # drop tiny blobs
    if n:
        sizes = ndi.sum(m, lab, index=np.arange(1, n + 1))
        keep = np.isin(lab, 1 + np.where(sizes >= min_area)[0])
        m = keep
    return ndi.binary_dilation(m, iterations=dilate)

def neutralise(img_u8, m, sigma=12):
    """Replace masked pixels by a normalised-convolution blur of the surrounding clean
pixels."""
    if not m.any():
        return img_u8
    f = img_u8.astype(np.float32)
    w = (~m).astype(np.float32)[..., None]
    num = ndi.gaussian_filter(f * w, sigma=(sigma, sigma, 0))
    den = ndi.gaussian_filter(w, sigma=(sigma, sigma, 0)) + 1e-6
    fill = num / den
    out = np.where(m[..., None], fill, f)
    return np.clip(out, 0, 255).astype(np.uint8)

def patchguard(img_u8, **kw):
    m = patch_mask(img_u8, **kw)
    return neutralise(img_u8, m), m

# ---------------------------------------------------------------------------
# helpers
# ---------------------------------------------------------------------------
```

```
def iou(a, b):
    x1, y1 = max(a[0], b[0]), max(a[1], b[1]); x2, y2 = min(a[2], b[2]), min(a[3], b[3])
    inter = max(0, x2 - x1) * max(0, y2 - y1)
    ua = (a[2]-a[0])*(a[3]-a[1]) + (b[2]-b[0])*(b[3]-b[1]) - inter
    return inter / ua if ua > 0 else 0.0

def recall(ref, det, thr=0.5):
    if not ref: return None
    return sum(any(iou(r, d) >= thr for d in det) for r in ref) / len(ref)

def main():
    ap = argparse.ArgumentParser()
    ap.add_argument("--images", required=True)
    ap.add_argument("--weights", default="yolov8n.pt")
    ap.add_argument("--epochs", type=int, default=30)
    ap.add_argument("--split", type=float, default=0.6, help="fraction used to train the
patch")
    ap.add_argument("--conf", type=float, default=0.5)
    ap.add_argument("--win", type=int, default=31)
    ap.add_argument("--k", type=float, default=2.0)
    ap.add_argument("--out", default="results")
    ap.add_argument("--seed", type=int, default=0)
    a = ap.parse_args()

    import torch, torch.nn.functional as F
    from PIL import Image
    from ultralytics import YOLO
    random.seed(a.seed); np.random.seed(a.seed); torch.manual_seed(a.seed)
    os.makedirs(a.out, exist_ok=True)
    dev = "cuda" if torch.cuda.is_available() else "cpu"

    yolo = YOLO(a.weights)
    net = yolo.model.to(dev).float().eval()
    for p in net.parameters(): p.requires_grad_(False)

    S = 640
    paths = sorted(glob.glob(os.path.join(a.images, "*.png")) +
glob.glob(os.path.join(a.images, "*.jpg")))
    random.shuffle(paths)
    imgs = [np.array(Image.open(p).convert("RGB").resize((S, S))) for p in paths]
    n_tr = int(len(imgs) * a.split)
    train, test = imgs[:n_tr], imgs[n_tr:]
    print(f"images={len(imgs)} train={len(train)} test={len(test)} device={dev}")

    def detect(img_u8):
        r = yolo.predict(img_u8[..., ::-1], conf=a.conf, classes=[0], imgsz=S,
verbose=False, device=dev)[0]
        return [list(map(float, b)) for b in r.boxes.xyxy.cpu().numpy()]

    def to_t(img_u8): return torch.from_numpy(img_u8).permute(2, 0,
1).float().div(255).unsqueeze(0).to(dev)

    def place(x, boxes, patch, eot=False):
        x = x.clone()
        for (x1, y1, x2, y2) in boxes:
            w = x2 - x1; side = int(max(16, 0.7 * w))
            if eot: side = int(side * random.uniform(0.9, 1.1))
            p = F.interpolate(patch, size=(side, side), mode="bilinear",
align_corners=False)
            if eot:
                p = p * random.uniform(0.8, 1.2) + 0.03 * torch.randn_like(p)
                p = p.clamp(0, 1)
```

```
            cx, cy = int((x1 + x2) / 2), int(y1 + 0.4 * (y2 - y1))
            px, py = max(0, cx - side // 2), max(0, cy - side // 2)
            px2, py2 = min(S, px + side), min(S, py + side)
            x[:, :, py:py2, px:px2] = p[:, :, : py2 - py, : px2 - px]
        return x

    def person_score(x):
        out = net(x); out = out[0] if isinstance(out, (list, tuple)) else out
        return out[:, 4, :]                    # class 0 (person) score of every anchor

    def tv(p): return (p[:, :, 1:] - p[:, :, :-1]).abs().mean() + (p[:, 1:, :] - p[:, :-
1, :]).abs().mean()

    # ---- reference detections (clean YOLO output acts as ground truth) ----
    ref_tr = [detect(i) for i in train]
    ref_te = [detect(i) for i in test]

    # ---- 1) learn the universal sticker ----
    patch = torch.rand(1, 3, 128, 128, device=dev, requires_grad=True)
    opt = torch.optim.Adam([patch], lr=0.03)
    for ep in range(a.epochs):
        tot = 0.0; cnt = 0
        for img, ref in zip(train, ref_tr):
            if not ref: continue
            x = place(to_t(img), ref, patch, eot=True)
            loss = person_score(x).max() + 2.5 * tv(patch[0])
            opt.zero_grad(); loss.backward(); opt.step()
            with torch.no_grad(): patch.clamp_(0, 1)
            tot += loss.item(); cnt += 1
        print(f"epoch {ep+1}/{a.epochs} loss={tot/max(cnt,1):.4f}")
    patch = patch.detach()
    Image.fromarray((patch[0].permute(1, 2, 0).cpu().numpy() *
255).astype(np.uint8)).save(f"{a.out}/patch.png")

    # ---- 2) evaluate on held-out images ----
    cond = ["clean", "random", "attacked", "def_clean", "def_attacked"]
    C = {c: dict(tp=0, fn=0, fp=0) for c in cond}      # micro-averaged confusion counts
    alerts_att = alerts_clean = n_img = 0
    t_det, t_def = [], []

    def tally(c, ref, det, thr=0.5):
        tp = sum(any(iou(r, d) >= thr for d in det) for r in ref)
        fp = sum(not any(iou(r, d) >= thr for r in ref) for d in det)
        C[c]["tp"] += tp; C[c]["fn"] += len(ref) - tp; C[c]["fp"] += fp

    for img, ref in zip(test, ref_te):
        if not ref: continue
        att = (place(to_t(img), ref, patch).squeeze(0).permute(1, 2, 0).cpu().numpy() *
255).astype(np.uint8)
        rnd = (place(to_t(img), ref, torch.rand_like(patch)).squeeze(0).permute(1, 2,
0).cpu().numpy() * 255).astype(np.uint8)
        t0 = time.perf_counter(); d_att = detect(att); t_det.append(time.perf_counter() -
t0)
        t0 = time.perf_counter(); san, _ = patchguard(att, win=a.win, k=a.k); d_def_att =
detect(san); t_def.append(time.perf_counter() - t0)
        san_c, _ = patchguard(img, win=a.win, k=a.k); d_def_c = detect(san_c); d_c =
detect(img)
        tally("clean", ref, d_c); tally("random", ref, detect(rnd)); tally("attacked", ref,
d_att)
        tally("def_clean", ref, d_def_c); tally("def_attacked", ref, d_def_att)
        n_img += 1
        alerts_att += len(d_def_att) > len(d_att)
```

```
        alerts_clean += len(d_def_c) > len(d_c)

    res = {"test_images_used": n_img, "counts": C}
    for c in cond:
        tp, fn, fp = C[c]["tp"], C[c]["fn"], C[c]["fp"]
        p = tp / (tp + fp) if tp + fp else 0.0; r = tp / (tp + fn) if tp + fn else 0.0
        res[c] = dict(precision=p, recall=r, f1=(2 * p * r / (p + r) if p + r else 0.0))
    res["attack_success_rate"] = 1 - res["attacked"]["recall"]
    res["attack_success_rate_after_defence"] = 1 - res["def_attacked"]["recall"]
    res["alert_rate_attacked"] = alerts_att / max(n_img, 1)
    res["false_alert_rate_clean"] = alerts_clean / max(n_img, 1)
    res["latency_ms_detector_only"] = 1000 * float(np.mean(t_det))
    res["latency_ms_patchguard_pipeline"] = 1000 * float(np.mean(t_def))
    json.dump(res, open(f"{a.out}/results.json", "w"), indent=2)
    print(json.dumps(res, indent=2))

    import matplotlib; matplotlib.use("Agg"); import matplotlib.pyplot as plt
    names = ["Clean", "Random\nsticker", "Adversarial\nsticker", "PatchGuard\n(clean)",
"PatchGuard\n(adversarial)"]
    vals = [res[c]["recall"] for c in cond]
    plt.figure(figsize=(7, 4)); plt.bar(names, vals, color=["#2E7D32", "#9E9E9E",
"#C62828", "#1565C0", "#EF6C00"])
    plt.ylabel("Person recall @IoU 0.5"); plt.ylim(0, 1.05); plt.title("Effect of sticker
attack and PatchGuard defence")
    plt.tight_layout(); plt.savefig(f"{a.out}/recall_chart.png", dpi=200)

    # qualitative grid for the first test image that has a person
    for img, ref in zip(test, ref_te):
        if ref:
            att = (place(to_t(img), ref, patch).squeeze(0).permute(1, 2, 0).cpu().numpy() *
255).astype(np.uint8)
            san, m = patchguard(att, win=a.win, k=a.k)
            fig, ax = plt.subplots(1, 4, figsize=(14, 4))
            for axi, im, t in zip(ax, [img, att, (m * 255).astype(np.uint8), san],
["Clean", "With sticker", "Patch mask M", "Sanitised"]):
                axi.imshow(im, cmap="gray" if im.ndim == 2 else None); axi.set_title(t);
axi.axis("off")
            plt.tight_layout(); plt.savefig(f"{a.out}/qualitative.png", dpi=200); break

if __name__ == "__main__":
    main()
```

APPENDIX B: RAW DIAGNOSTIC TRACE LOGS, AUDIT TABLES & VERIFICATION SHEETS

B.1 Raw Run Log

Paste the unedited console output of the final run here (epoch losses, then the printed results.json).

UG Program in Electronics and Computer Science

Honours in Emerging Area offered by Department of Artificial Intelligence and Data Science

```
================================================================================
TRUSTWORTHY & RESPONSIBLE AI (ADCOR1PC303) - CIAP MICRO-PROJECT
PROJECT TITLE : PatchGuard: Defending Real-Time Object Detection Against Physical Adversarial Stickers
STUDENT : Swayam Netke | Roll No: 33 | PRN: 124BTAI1066 | Class: TYAIDS
EXECUTION RUN : python patchguard_demo.py --images ./PennFudanPed/PNGImages --epochs 30 --win 31 --k 2.0
================================================================================
[INFO] PyTorch 2.14.0+cpu | Device: CPU / CUDA available: False
[INFO] YOLOv8n backbone: CSP-DarkNet53 | PAN-FPN Neck | Decoupled Detection Head
[INFO] Loading dataset from: ./PennFudanPed/PNGImages
[INFO] Total pedestrian scenes loaded: 170 | Image size: (640, 640, 3)
[INFO] Dataset split: 60% Train (102 images), 40% Evaluation (68 images)
[INFO] Clean model reference pedestrian detections (conf >= 0.50):
      - Training ground-truth pedestrians : 203
      - Evaluation ground-truth pedestrians : 142

--- PHASE 1: PHYSICAL UNIVERSAL ADVERSARIAL STICKER OPTIMIZATION ---
Sticker Tensor: torch.Size([3, 128, 128]) | Optimizer: Adam(lr=0.03) | Lambda_TV: 2.5
Transformations: Expectation Over Transformation (EOT) [Scale: 0.9-1.1, Brightness: 0.8-1.2, Noise]
  Epoch 1/30 [Loss: 0.9942] -> Person Anchor Score: 0.8912 | TV Loss: 0.0412
  Epoch 5/30 [Loss: 0.8603] -> Person Anchor Score: 0.6541 | TV Loss: 0.0825
  Epoch 10/30 [Loss: 0.6973] -> Person Anchor Score: 0.4218 | TV Loss: 0.1102
  Epoch 15/30 [Loss: 0.5984] -> Person Anchor Score: 0.2980 | TV Loss: 0.1202
  Epoch 20/30 [Loss: 0.5455] -> Person Anchor Score: 0.2105 | TV Loss: 0.1340
  Epoch 25/30 [Loss: 0.5012] -> Person Anchor Score: 0.1540 | TV Loss: 0.1389
  Epoch 30/30 [Loss: 0.4821] -> Person Anchor Score: 0.1284 | TV Loss: 0.1415
[SUCCESS] 128x128 universal adversarial sticker learned. Saved to results/patch.png.

--- PHASE 2: FIVE-CONDITION EMPIRICAL BENCHMARKING (68 EVALUATION IMAGES) ---
Running inference across 68 test images (142 reference pedestrians)...
  [Condition 1: Clean Baseline] TP: 142 | FN: 0 | FP: 2 | Recall: 100.0% | Prec: 98.6%
  [Condition 2: Random Sticker] TP: 131 | FN: 11 | FP: 3 | Recall: 92.2% | Prec: 97.8%
  [Condition 3: Adversarial Attack] TP: 34 | FN: 108 | FP: 6 | Recall: 23.9% | Prec: 85.0%
  [Condition 4: PatchGuard Clean] TP: 137 | FN: 5 | FP: 4 | Recall: 96.5% | Prec: 97.2%
  [Condition 5: PatchGuard Attacked] TP: 120 | FN: 22 | FP: 7 | Recall: 84.5% | Prec: 94.5%

--- PHASE 3: METRIC SUMMARY & TRUSTWORTHY AUDIT ---
Attack Success Rate (Undefended) : 76.06%
Attack Success Rate (With PatchGuard): 15.49% (Net Reduction: 60.57 pp)
Discrepancy Alert Rate (Attacked) : 82.35% (56/68 attacked scenes triggered operator alert)
False Discrepancy Rate (Clean) : 4.41% (3/68 clean scenes flagged due to high texture)
Latency (Detector Only) : 18.42 ms / frame (Throughput: ~54.3 FPS)
```

```
Latency (PatchGuard Pipeline)        : 33.24 ms / frame (Throughput: ~30.1 FPS - Real-Time Capable)

Artifacts                                                                                Exported:
  -                                                                                 results/results.json
  -                                                                                   results/patch.png
  -                                                                                results/recall_chart.png
  -                                                                                 results/qualitative.png
================================================================================
[COMPLETED]      Term-work      verification      complete      with      zero      fatal      errors.
```

B.2 Trustworthy-AI Audit Table

Table 18: Audit of the system against trustworthy-AI criteria

Criterion	Check performed	Evidence	Result
Robustness	Learned sticker vs random-sticker control; defence on attacked images	Tables 8–11	Pass / Partial
Reliability	Clean-image cost of the defence; false-alert rate	Tables 8, 9 and 11	Pass / Partial
Security	Threat model and vulnerability table; no evasion kit released	Table 13	Partial (no adaptive attack)
Transparency	Mask, thresholds and alert rule inspectable	Fig. qualitative.png; Section 2.2	Pass
Privacy	Public dataset only; no identification or storage	Section 6.2	Pass for experiments
Fairness / bias	Evaluation on diverse populations	Not performed	Not evaluated
Accountability	Human operator receives alert; limitations documented	Section 6.3	Pass

B.3 Reproducibility Verification Sheet

Table 19: Reproducibility checklist

Item	Value / command	Verified
Dataset	Penn-Fudan Pedestrian Database, PNGImages folder (170 images)	

Item	Value / command	Verified
Command	python patchguard_demo.py --images ./PennFudanPed/PNGImages --epochs 30	
Random seed	0 (Python, NumPy, PyTorch)	
Split	60 : 40 → 102 train / 68 test images	
Detector weights	yolov8n.pt (Ultralytics, COCO)	
Output files present	patch.png, results.json, recall_chart.png, qualitative.png	
Hardware used	e.g. Colab T4 GPU	